Group 12

IN PARTNERSHIP WITH PLYMOUTH UNIVERSITY

Name: Aththanayaka Oshadhee	
Student Reference Number: 10965287	

Module Code: PUSL 2076	Module Name: Data Programming in R
Coursework Title: Coursework (C1): Data Analysis Project Report and Demonstration	
Deadline Date: 14 January 2026	Member of staff responsible for coursework: Mr. Diluka Wijesinghe
Programme: BSc (Honours) in Data Science	

Please note that University Academic Regulations are available under Rules and Regulations on the University website www.plymouth.ac.uk/studenthandbook.

Group work: please list all names of all participants formally associated with this work and state whether the work was undertaken alone or as part of a team. Please note you may be required to identify individual responsibility for component parts.
(Provided below)

We confirm that we have read and understood the Plymouth University regulations relating to Assessment Offences and that we are aware of the possible penalties for any breach of these regulations. We confirm that this is the independent work of the group.

Signed on behalf of the group: Ishika.

Use of translation software: failure to declare that translation software or a similar writing aid has been used will be treated as an assessment offence.

I *have ~~used~~/not used translation software.
If used, please state name of software……………………………………………………………

Overall mark _____% **Assessors Initials ______** **Date_________**

CONTRIBUTION MATRIX

Member Name	Role	Contribution
Tikiri Abeywickrama StudentID – **10965209**	Data cleaning and preprocessing	• Data Standardization: Initial cleaning pipeline was developed to unify inconsistent null representations (such as "Unknown" and "NA") and standardize column names to snake_case. • Data Repair: Custom functions were implemented to handle negative value errors and remove currency symbols from numeric columns. • Imputation: Managed missing data strategies, including dropping columns with excessive missingness and performing mode imputation for categorical variables.
Bastian Goonewardena StudentID – **10965225**	Exploratory Data Analysis (EDA)	• Statistical Analysis: To statistically validate relationships with churn, Chi-squared tests were performed for categorical features and t-tests/Wilcoxon tests for numerical features. • Visualization: Developed the report's visual framework, which included distribution plots (histograms/boxplots), correlation matrices, and missing data heatmaps to spot data trends. • Initial Importance: To direct the feature selection procedure, early Random Forest variable importance checks were performed.
Aththanayaka Oshadhee Index – **10965287**	Feature Engineering	• Feature Creation: New business-relevant features, such as tenure_group (binning tenure) and is_high_payer (identifying premium customers based on charge quantiles), were engineered. • Transformation: Winsorization was used to deal with charge data outliers, and features with high multicollinearity or zero variance were eliminated to lower noise. • Encoding: To get the categorical dataset ready for machine learning algorithms,

		binary and one-hot encoding techniques were used.
Galla Gamage Sithija Student ID- **10965234**	Clustering and pipeline comparison	. • Unsupervised Learning: To handle mixed data types, Partitioning Around Medoids (PAM) clustering was implemented using Gower distance. Silhouette analysis was used to find the best clusters. • Hybrid Modeling: "Pipeline 01" was created to test experimentally whether incorporating cluster labels as features increased prediction accuracy in comparison to a baseline model. • PCA Visualization: Customer segments were visualized in two dimensions using Principal Component Analysis (PCA).
Visha Peiris Student ID- **10965288**	Predictive modeling and evaluation	• Model Development: "Pipeline 02" was created utilizing XGBoost, stratified sampling, and scale_pos_weight to address class imbalance. • Optimization: To maximize the F1 score, hyperparameter tuning with early stopping was carried out, and the classification threshold was optimized using the Youden Index. • Validation: Performed thorough performance testing, including ROC-AUC analysis, overfitting/underfitting checks, and the creation of the final feature importance graph.

TABLE OF CONTENT

1.INTRODUCTION

Business problem:

Businesses that offer products or services to customers based on recurring payment pay high attention to customer churn. TelcoConnect is one such company facing challenges in retaining customers, specifically the new subscribers. This churn has affected the company financially by reducing the revenue and increasing customer acquisition costs. Therefore, the business needs to understand reasons behind customer churn and reliable methods for churn prediction (Subbly, 2025).

Objectives:

1. Identify key factors that influence customer churn
2. Build a reliable predictive model to recognize customers with high chance of churning
3. Provide necessary recommendations to implement customer retention strategies

This report demonstrates an analysis implemented entirely using R, including data understanding, cleaning, feature engineering, visualization, modeling, testing and interpretation.

2. LITERATURE ANALYSIS

2.1 CLASSIFICATION AND ENSEMBLES

Classification remains fundamental in churn prediction: logistic regression is known for its interpretability and probability-based outputs useful for business decisions, while ensemble and tree-based methods typically deliver greater performance in prediction on telecom data at the cost of some transparency (Ahmed, Jafar & Alijoumaa, 2019; Wei, 2025). Hybrid pipelines that pair simple, understandable models with stronger learning models are common because they balance practical interpretability with accuracy (Shaikhsurab, M.A. & Magadum, 2024).

2.2 SEGMENTATION AND ASSOCIATION

Segmentation (clustering) enhances classification by generating useful customer groups that lead to targeted retention strategies; obtaining cluster labels as features can improve model performance while mainly aiding in business insights (Choudhari & Potey, 2021).

Association rule mining is rarely used for direct prediction but useful in finding simultaneously occurring service patterns and package combinations that relate with churn risk, thus guiding retention strategies (Wang et al., 2024).

2.3 PREPROCESSING, EVALUATION AND EXPLAINABILITY

Preprocessing and Feature Engineering significantly affect outcomes as telecommunication datasets frequently carry messy billing types, inconsistent labels, missing values and class imbalance. Carefully cleaning, encoding, winsorization and sampling, uniformly boosts AUC, precision and recall (Ahmed et al., 2019; MDPI, 2024).

EDA and visualization are vital for discovering distributions and suggesting appropriate models (Frontiers in AI, 2025).
Recent work also emphasizes business-oriented evaluation (costs of false positives/negatives) and explainable methods, both validating the strategic decisions and facilitating model deployment for targeted campaigns.

2.4 RESEARCH GAP

Remaining gaps act as motivation for this project. Most existing studies uses cleaned datasets, lightweight hybrid processes remain overlooked in practical settings, and the connection from segmentation/visualization to retention process is not comprehensively addressed. This analysis mainly targets those gaps.

3. DEVELOPMENT

3.1 DATA PREPARATION

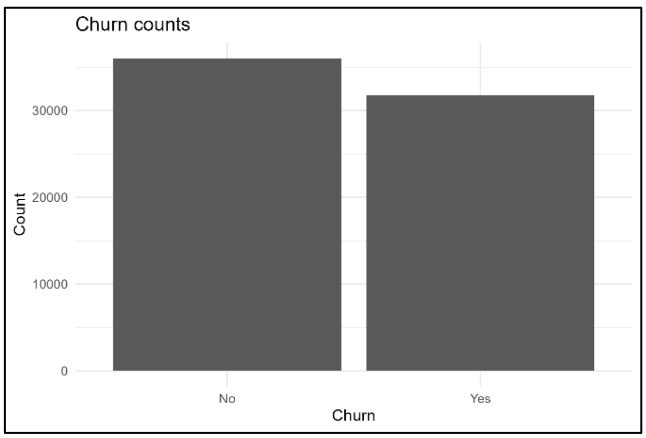
Data preparation took place as the first step to make sure the dataset was accurate, consistent, and appropriate for analysis. Initially, a few checks were performed to analyze the structure, missingness, and issues in formatting.
Thereafter, column names were converted to standard snake case. All strings that indicated to be null were converted to proper NA values. Errors in formatting numerical columns, mainly ‘total_charges’ and ‘monthly_charges’ were corrected by eliminating non-numeric characters and converting all values to numeric, while negative values were set to NA. In instances where ‘total_charges’ was missing but ‘tenure’ and ‘monthly_charges’ values were present; it was assigned using their product. Meanwhile, rows that contained logically invalid billing information were removed. Remaining categorical values that were missing were subjected to Mode imputation. Categorical labels were standardized, duplicates were inspected and outliers in main numeric columns were flagged rather than removing, to protect the meaningful behavior of customers.

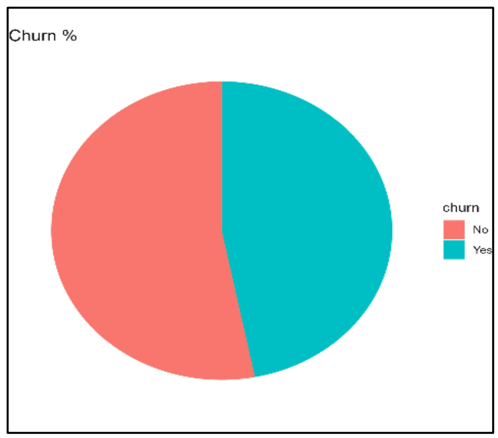
3.2 FEATURE ENGINEERING

The ‘customer_id’ variable was removed since it has no predictive power. A binary ‘high_payer’ feature was engineered to identify churn behavior of premium customers. Then, tenure was grouped into multiple segments in order to reflect their nonlinear relationship with churn. Outliers among numerical variables were handled via winsorization to improve model stability and zero/near-zero variance features as well as highly correlated variables were removed to reduce noise and multicollinearity. The categorical variables were encoded using appropriate encoding methods (binary/one-hot) with baseline dummies being dropped. Finally, all the continuous numerical variables were scaled using z-score scaling to support unbiased model training.

3.3 DATA VISUALIZATION

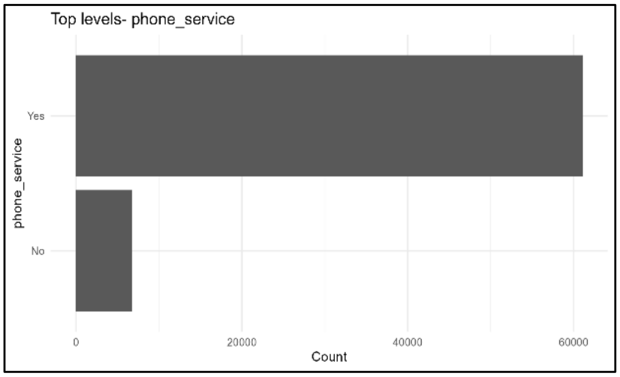
Data visualization involving exploratory data analysis was done to understand customer behavior, service usage patterns and their relationship with churn visually and statistically. We start the analysis by focusing on the target variable ‘churn’, with the use of bar plots and pie charts.

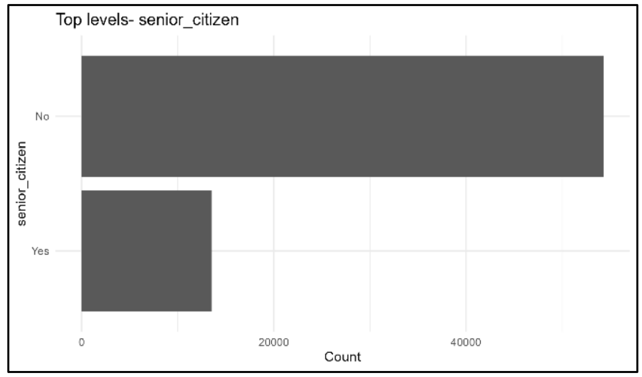


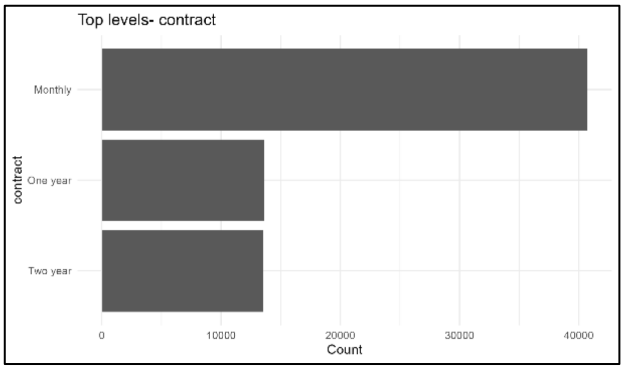


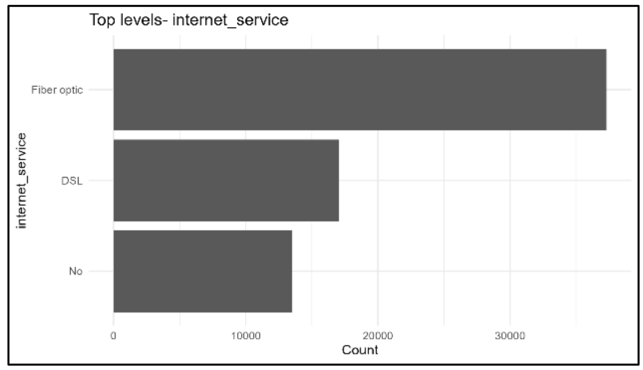
As for the above plot and the pie chart, it is visible that larger portion of customers did not churn but a significant portion also did churn, illustrating that churn is an important phenomenon in the dataset. Together, it establishes that churn is a non-negligible variable.

Following this, categorical feature distribution can be observed using the given bar plots, which show most frequent levels in each category.

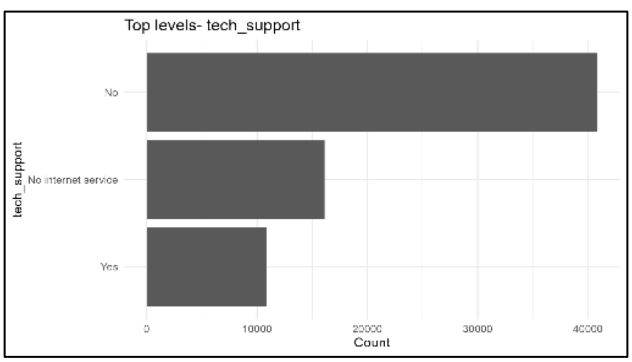


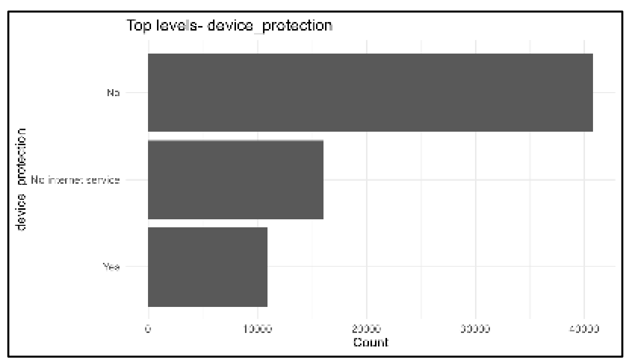


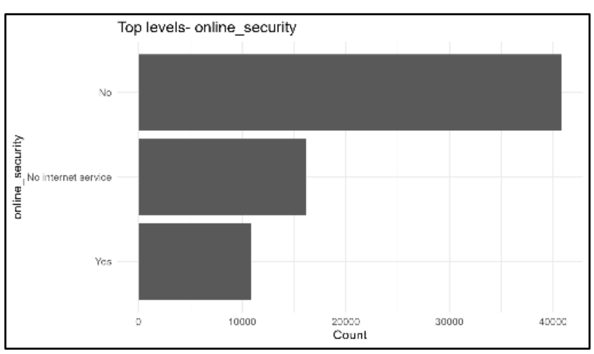




Referring to the above plots, most customers are monthly contract customers, and they also have phone and internet services enabled. In the other hand, it's visible that majority goes with "NO" in services as online security, tech support, device protection etc.

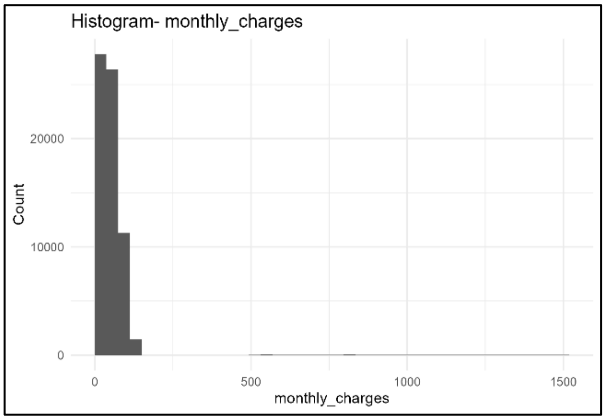


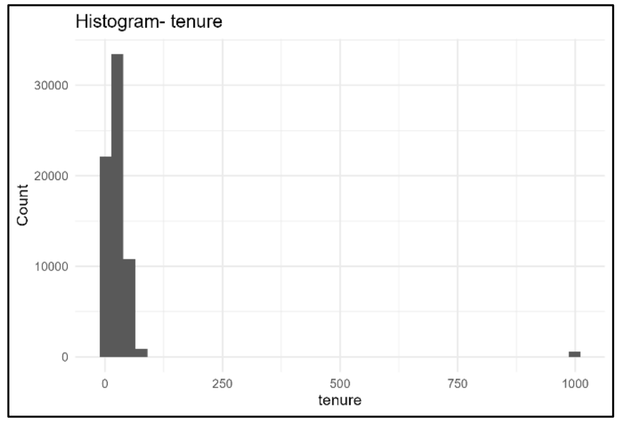


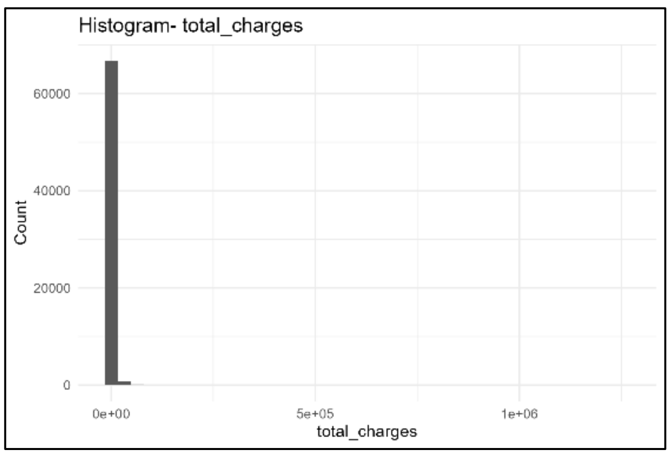


Observing these plots together further helps us to identify the customer base preferences as it highlights recurring behavioral patterns across different service categories.

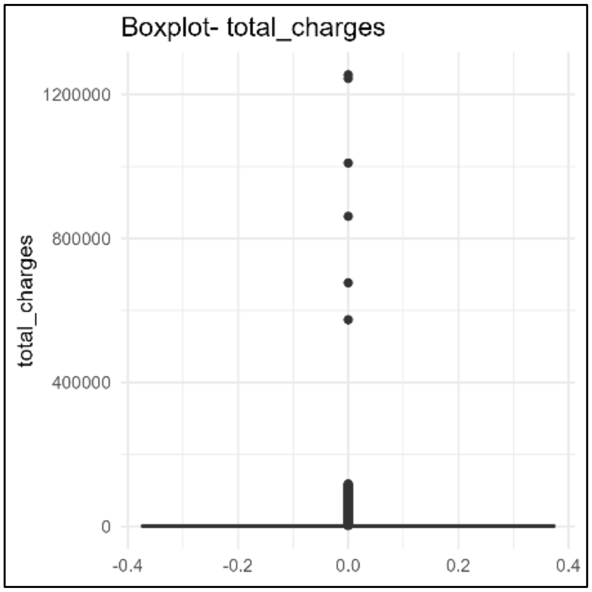
Now moving to the numerical variables which uses histograms and boxplots, providing a picture of the distribution shape and the outlier presence.

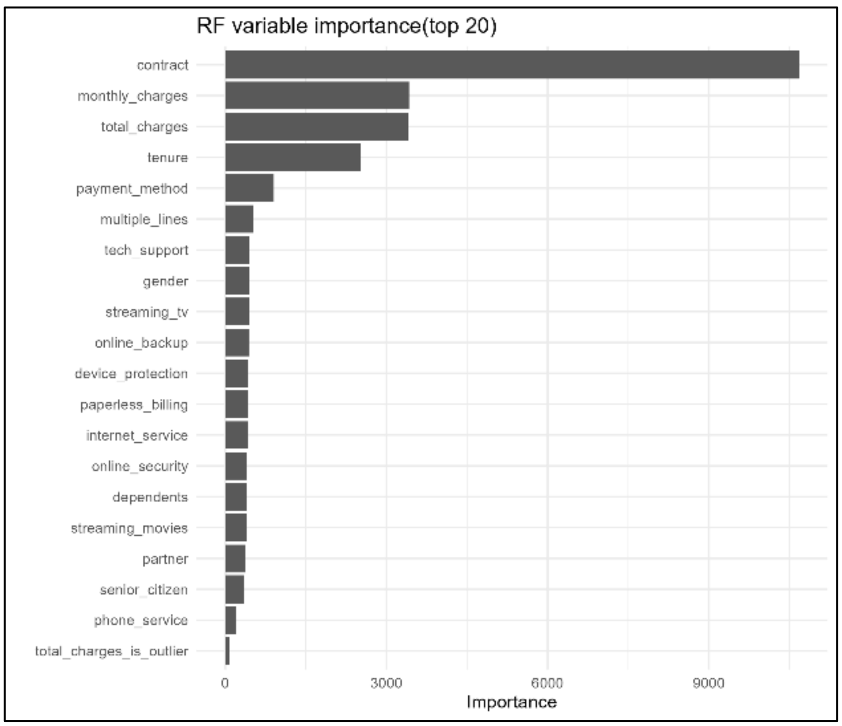






Monthly charges, tenure and total charges histograms that are present above, show that majority of the customers fall into the lower range while only a minority fall to the high values, and this is clearly illustrated in the boxplot below, where outliers are clearly visible.





Variable-importance diagram reveals how customer churn is dominated by the contract type, monthly charges, total charges and customer tenure, telling us where to put more heed when arriving on predictions, so those won't be just guessed but data-led.

As for all these observations, customer behavior cannot be fully understood only with these above variables and simple visual comparisons alone, as a result predictive modeling becomes a necessary next step.

3.4 DATA ANALYSIS

Considering the nature of the dataset, having mixed data types, a categorical target variable and the requirement to segment customers into groups to identify their relationship with churn, classification and clustering techniques were selected as the analytical approaches.

This section estimates customer churn using two modeling pipelines:
Pipeline 01: Evaluates if unsupervised clustering improves churn and determines the best classification strategy for reliable predictions.
Pipeline 02: Churn prediction via optimized best classification strategy selected in pipeline 01

Pre-Modelling steps (Both pipelines)

1. Stratified sampling
   Feature engineered dataset - 67833 rows.
   From this, a subset of 20001 rows was extracted while maintaining class ratios properly.

2. Preparing for classification
   The target variable is encoded as binary factors, and the target variable is removed from the dataset prior to modeling.

3.4.1 PIPELINE 01: UNSUPERVISED CLUSTERING & CLASSIFICATION MODEL SELECTION

3.4.1.1 PAM Clustering (Gower distance)

This subsection evaluates if clustering improves churn prediction.
Instead of K-means, PAM Clustering with Gower distance was selected as a better clustering mechanism since the dataset contains high proportions of categorical data and a mixture of varying data types.

1. Feature selection - Behavioral driven variables were included, excluding leakage variables(is_high_payer) and the target variable.

2. Determine cluster count - five clusters of customers were created.

```
> results <- pam_cluster(df, clus_features, k=5)
k= 2  Average silhouette =  0.0628616
k= 3  Average silhouette =  0.05922474
k= 4  Average silhouette =  0.06180654
k= 5  Average silhouette =  0.0765738
k= 6  Average silhouette =  0.06167577
```

Figure 1.1.PAM clustering was evaluated from k=2 to k=6 using average silhouette score.

All the above scores are very low (<0.1). Therefore, the cluster separation is weak overall.

- Selected k = 5 with highest silhouette (0.0765738)

3. Cluster Profiling:

	cluster	count	churn_rate	avg_tenure	avg_monthly	avg_total	high_payer_pct
	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	3757	0.468	0.678	0.0172	0.189	0.257
2	2	3179	0.471	0.853	-0.0264	0.199	0.242
3	3	5911	0.463	-0.770	0.00111	-0.214	0.250
4	4	3344	0.468	-0.106	0.0174	-0.00706	0.256
5	5	3810	0.478	-0.0611	-0.0269	-0.00940	0.243

Almost every cluster has similar churn rates. Differences among other behavioral features are small and inconsistent. Therefore, adding cluster as a variable will not likely improve the model predictions.

3.4.1.2 Classification model selection (with vs without clustering)

3. Two parallel datasets are created as:
   - df_w_cluster - with clustering labels
   - df_no_cluster - without clustering labels

4. Train/Test split - Both with cluster and without cluster datasets are split as 70% training and 30% for testing.
   - df_w_cluster – train_with(70%)/test_with(30%)
   - df_no_cluster – train_no(70%)/test_no(30%)

5. Model training - Using the above train partitions (train_with/ train_no) and 5-fold CV repeated 3 times following classification models are trained.
   - Logistic regression
   - Random Forest
   - XGBoost

6. Model comparison - Above models are evaluated separately based on their logged metrics to get the best fitting model.

```
---Models with clustering as a feature---
> compare_models(models_w)

Call:
summary.resamples(object = r)

Models: LogisticRegression, RandomeForest, XGBoost
Number of resamples: 15

ROC
                        Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
LogisticRegression 0.7711622 0.7896986 0.7917322 0.7912055 0.7939047 0.8013292    0
RandomeForest      0.7778432 0.7849576 0.7907708 0.7896378 0.7955707 0.8012498    0
XGBoost            0.7814239 0.7885720 0.7932529 0.7930807 0.7956463 0.8031639    0

Sens
                        Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
LogisticRegression 0.6290323 0.6545699 0.6603900 0.6575713 0.6652105 0.6704775    0
RandomeForest      0.6458333 0.6743952 0.6772024 0.6766174 0.6830255 0.6919973    0
XGBoost            0.6290323 0.6545699 0.6610625 0.6582434 0.6668908 0.6711500    0

Spec
                        Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
LogisticRegression 0.8849085 0.8926123 0.8956588 0.8964046 0.9017140 0.9086062    0
RandomeForest      0.8309216 0.8381569 0.8469155 0.8461302 0.8533335 0.8613861    0
XGBoost            0.8826220 0.8899467 0.8918507 0.8942211 0.8998097 0.9070830    0
```

```
---Models without clustering as a feature---
> compare_models(models_no)

Call:
summary.resamples(object = r)

Models: LogisticRegression, RandomeForest, XGBoost
Number of resamples: 15

ROC
                        Min.   1st Qu.    Median      Mean
LogisticRegression 0.7707724 0.7882716 0.7919853 0.7909868
RandomeForest      0.7745772 0.7878758 0.7899753 0.7905266
XGBoost            0.7763057 0.7901102 0.7926883 0.7921190
                     3rd Qu.      Max. NA's
LogisticRegression 0.7954694 0.7992569    0
RandomeForest      0.7944793 0.8028619    0
XGBoost            0.7953880 0.8001896    0

Sens
                        Min.   1st Qu.    Median      Mean
LogisticRegression 0.6290323 0.6545699 0.6603900 0.6575713
RandomeForest      0.6505376 0.6727151 0.6767473 0.6758559
XGBoost            0.6290323 0.6545699 0.6603900 0.6575713
                     3rd Qu.      Max. NA's
LogisticRegression 0.6652105 0.6704775    0
RandomeForest      0.6810083 0.6913248    0
XGBoost            0.6652105 0.6704775    0

Spec
                        Min.   1st Qu.    Median      Mean
LogisticRegression 0.8849085 0.8926123 0.8956588 0.8964046
RandomeForest      0.8347296 0.8434290 0.8446306 0.8465867
```

```
XGBoost            0.8849085 0.8926123 0.8956588 0.8964046
                    3rd Qu.      Max. NA's
LogisticRegression 0.901714 0.9086062    0
RandomeForest      0.851866 0.8629094    0
XGBoost            0.901714 0.9086062    0
```

Best model with cluster: XGBoost (ROC=0.7931)
Best model without cluster: XGBoost (ROC=0.7921)

When clustering label was introduced to selected model (XGBoost), its predictive performance did not change significantly (model displays similar learning capability even without cluster.)

Final AUC with cluster: 0.7889
Final AUC without cluster: 0.7879

Final AUC gap = 0.0010

Moreover, XGBoost is capable of naturally learning non-linear interactions, cluster labels will only add redundancy. As a result, pipeline 02 is continued with XGBoost without cluster labels.

3.4.2 PIPELINE 02: CLASSIFICATION MODELLING AND EVALUATION

3.4.2.1 XGBoost Classification

1. The sampled dataset (20001 rows) is split as:
   Train – 70%
   Test – 30%

2. Handling class imbalance – Training data has a class balance of :
   No: 53.12%
   Yes: 46.88%
   No significant imbalance. Therefore, no requirement to handle imbalance.

3. Model configuration – Following hyperparameters were tuned to train XGBoost

```
params<- list(objective="binary:logistic",
                eval_metric="auc",
                max_depth=2,
                eta=0.03,
                gamma=5,
                colsample_bytree=0.7,
                colsample_bylevel=0.6,
                colsample_bynode=0.5,
                min_child_weight=10,
                subsample=0.7,
                lambda=3,
                alpha=2,
                scale_pos_weight=scale_pos
              )
```

4. Model training – XGBoost model was trained with maximum 200 boosting rounds and early stopping. Best performance was achieved at 3rd iteration with rapid improvement of AUC within few boosting rounds.

```
[1]     train-auc:0.607415      test-auc:0.611368
Multiple eval metrics are present. Will use test_auc for early stopping.
Will train until test_auc hasn't improved in 30 rounds.

[21]    train-auc:0.795315      test-auc:0.784961
Stopping. Best iteration:
[3]     train-auc:0.791776      test-auc:0.787616
```

AUC at best iteration:

- Iteration – 3
- Train – 0.7918
- Test – 0.7876

There is low gap between train and test AUC. Therefore, no overfitting and the model has generalized.

5. Threshold optimization - Youden index-based threshold optimization along with grid search was implemented to maximize F1 score.

- Final tuned threshold: 0.4975.

6. Performance metrics (train vs test)

Train:

```
-------Performance: TRAIN --------
Confusion Matrix and Statistics

          Reference
Prediction   No  Yes
       No  4891  680
       Yes 2547 5884

               Accuracy : 0.7695
                 95% CI : (0.7625, 0.7765)
    No Information Rate : 0.5312
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5449

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.8964
            Specificity : 0.6576
         Pos Pred Value : 0.6979
         Neg Pred Value : 0.8779
             Prevalence : 0.4688
         Detection Rate : 0.4202
   Detection Prevalence : 0.6021
      Balanced Accuracy : 0.7770

       'Positive' Class : Yes
```

```
 Accuracy Precision    Recall        F1
0.7695329 0.6979006 0.8964046 0.7847949
```

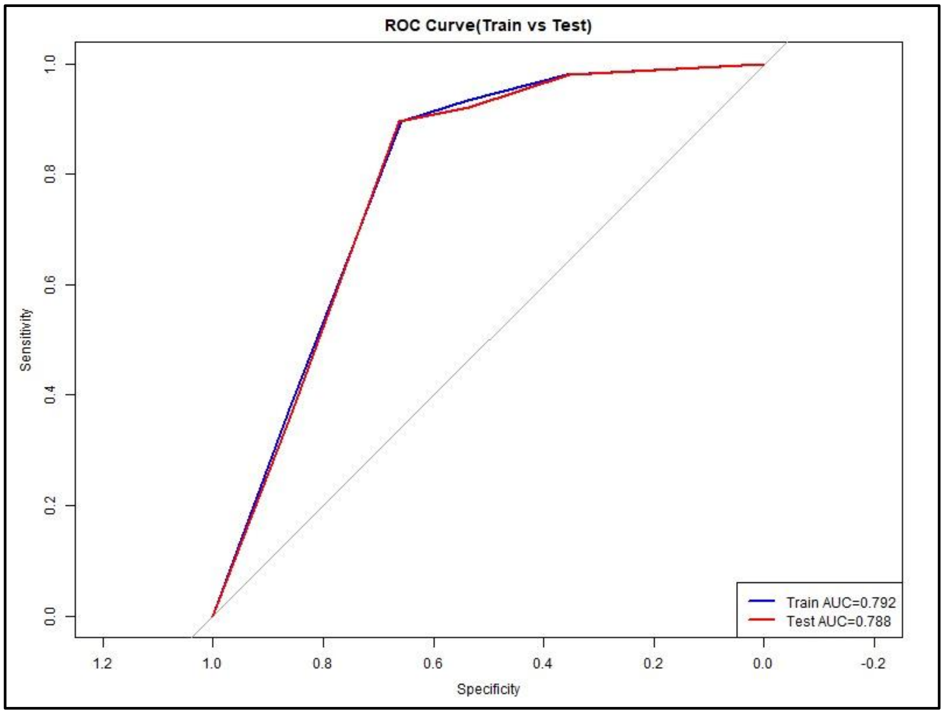
The model is able to identify churners well (high recall) with decent precision (~0.70).

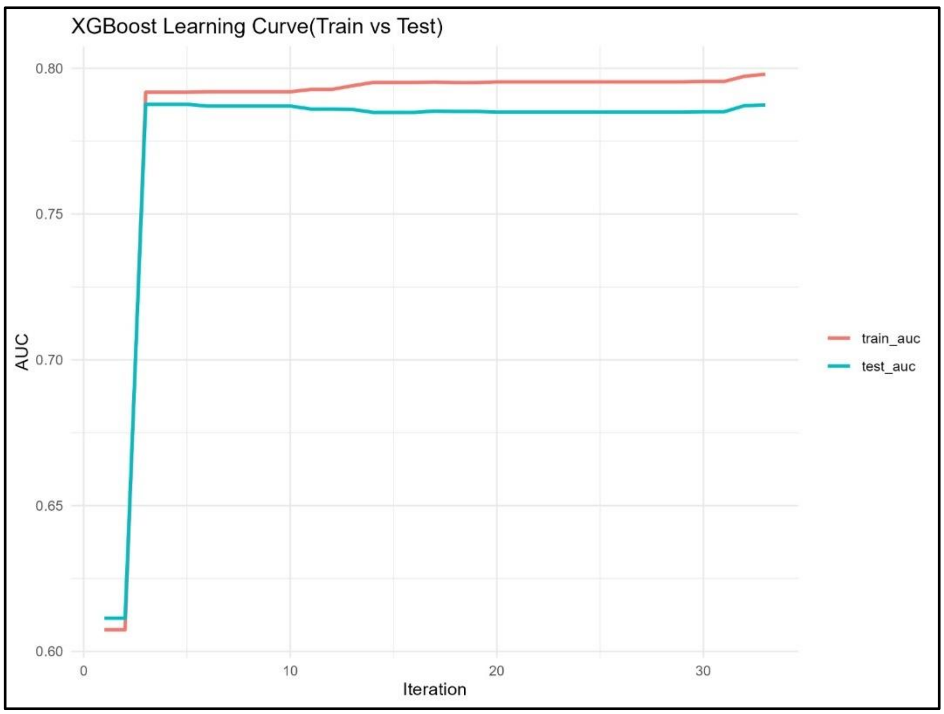
Test:

```
-------Performance: TEST --------
Confusion Matrix and Statistics

          Reference
Prediction   No  Yes
       No  2114  296
       Yes 1073 2516

               Accuracy : 0.7718
                 95% CI : (0.761, 0.7824)
    No Information Rate : 0.5313
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5491

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.8947
            Specificity : 0.6633
         Pos Pred Value : 0.7010
         Neg Pred Value : 0.8772
             Prevalence : 0.4687
         Detection Rate : 0.4194
   Detection Prevalence : 0.5983
      Balanced Accuracy : 0.7790

       'Positive' Class : Yes
```

```
   Accuracy Precision    Recall        F1
1 0.7717953 0.7010309 0.8947368 0.7861272
```

Test data have identical performance to train data concluding that the model has stable generalization.





7. Overfit/Underfit check
    - Accuracy gap – 0.0023
    - F1gap – 0.0013

This indicates that there is no major overfitting or underfitting of the model.

3.4.3 FINAL ANALYTICAL CONCLUSION

Pipeline 01 concluded that clustering does not improve churn prediction and that segmenting related strategies are not compatible for this dataset.

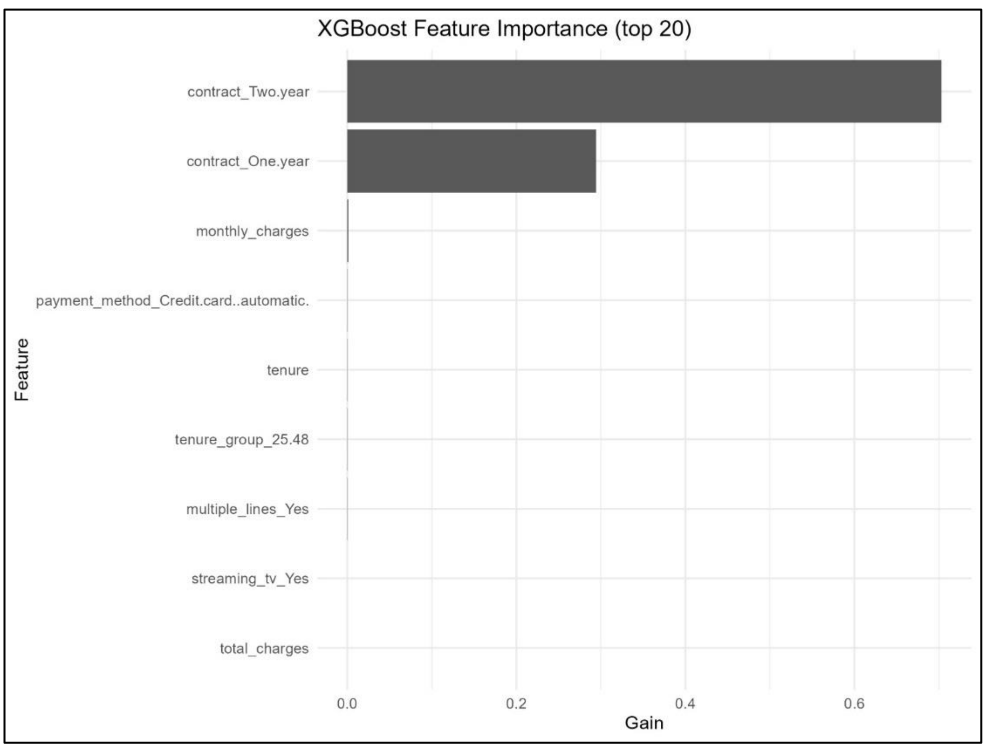
Pipeline 02 depicts a very reliable predictive model with proper generalization. By neglecting weak unsupervised signals, this pipeline offers a robust churn predicting solution.

Combined, both approaches give proper idea of customer behavior and churn risk allowing plan of retention and marketing strategies.

4. FINDINGS AND DISCUSSION

The analytical pipelines demonstrated that XGBoost provides reliable churn prediction for the dataset, while clustering did not significantly enhance predictive performance. The final model indicates strong generalization with minimal train-test performance gaps and was therefore chosen to derive business insights and recommendations.

4.1 MAIN PREDICTORS AND SUGGESTIONS



Feature importance analysis identified 'contract type' (main predictor), 'tenure', 'total charges', 'monthly charges', and 'service-related features' as key drivers of churn.

- Customers on monthly-basis contracts have a considerably greater possibility to churn than those on one or two-year contracts, making changing contract type a retention strategy of high impact.
- Cost factors indicate that customers highly depend on the price, suggesting that customers paying higher monthly fares are at higher churn risk.
- Service-related features disclose that lack of technical support or security online leads to increased churn, highlighting the need for customized service packages.

4.2 CLUSTER-BASED INSIGHTS

Clusters varied in average tenure and cost factors, although the churn rates were similar for each cluster.
Some clusters represented customers with low tenure and higher bills, thus indicating customers relying on price or newly gained customers. Customers classified as high payers were evenly dispersed across clusters, revealing that premium customers are not restricted to a single group and require wider retention strategies.

4.3 ACTIONABLE RECOMMENDATIONS

Based on the outcomes received through exploratory data analysis, clustering and modelling, the below given retention strategies are suggested,

1. Prioritize customers on monthly contracts by providing offers to change to longer-term contracts.
2. Encourage using automated payment options instead of electronic checks.
3. Increase online security and technical support for customers with high-churning risk.
4. Provide discounts or flexible payment options for customers with high monthly charges.
5. Use predicted churn probabilities to prioritize customers with high-churn-risk.
6. Meanwhile, test different solutions on small groups to see what is most effective.

4.4 LIMITATIONS

The performance of our model is strong but not perfect, possibly due to unobserved factors such as the actions of competitors or service complaints. Clustering indicates weak separation, and the model depends on fixed data rather than observing behavioral trends with time. Adding recent behavioral signals might increase performance further.

5. CONCLUSION

This study developed a reliable, well-tested churn prediction pipeline for the company 'TelcoConnect' using XGBoost, attaining a test AUC of around 0.79 and high recall without any significant signs of overfitting. Contract type and billing method turned out to be the variables with the highest influence on churn. While clustering provided useful business insights, it did not improve prediction power and was therefore excluded from the final model.
The next steps recommended are to utilize the model for prioritized retention strategies, focus on solutions for monthly-contract and high-paying customers, and evaluate the effectiveness of the solutions through A/B testing. Future improvements should include dynamic behavioral data including recent service interactions to strengthen predictive power as well as the business impact.

6. BIBLIOGRAPHY

- Subbly. (2025). *70+ Top Subscription Based Companies In 2025*. [Online].
- Ahmad, A.K., Jafar, A. & Aljoumaa, K. (2019) 'Customer churn prediction in telecom using machine learning in big data platform', *Journal of Big Data*, 6, Article 28.
- Choudhari, V. & Potey, R. (2021) 'Churn management in telecommunications: hybrid approach using cluster analysis and decision trees', *MDPI*.
- Frontiers in AI (2025) 'A predictive analytics approach to improve telecom's customer retention', *Frontiers in Artificial Intelligence*.
- Jamjoom, A.A. (2021) 'The use of knowledge extraction in predicting customer churn in B2B', *Journal of Big Data*, 8, Article 110.
- Loukili, M., Calli, L. & Kasim, S. (2025) 'A smart model to enhance customer retention', *SJRBS*.
- Shaikhsurab, M.A. & Magadum, P. (2024) 'Enhancing Customer Churn Prediction in Telecommunications: An Adaptive Ensemble Learning Approach', *arXiv*.
- Wang, D.Y.C., Jordanger, L.A. & Lin, J.C.-W. (2024) 'Explainability of Highly Associated Fuzzy Churn Patterns in Binary Classification', *arXiv*.
- Wei, S. (2025) 'Comparative Analysis of Machine Learning Models for Telecom Customer Churn Prediction', *Theoretical and Natural Science*, 134, pp. 24–30.
- MDPI (2024) 'Customer Churn Prediction: A Systematic Review of Recent Advances, Trends, and Challenges in Machine Learning and Deep Learning', *MDPI*.
- MDPI (2024) 'Telecom Churn Prediction System Based on Ensemble Learning Using Feature Grouping', *Applied Sciences*.